

Less to Lose:

How an earlier credential reshaped  
university choice and earnings

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**Abstract**

University is a costly investment made under uncertainty: students pay while enrolled, learn only gradually whether a degree suits them, and receive much of the return only if they finish. This paper asks how degree length shapes educational choices and earnings. I study the Bologna Process, which changed degrees in two ways. Where a field already had a bachelor's and a master's, it shortened the bachelor's to three years. Where a field had a single long degree, it split that degree into a bachelor's and a master's. Using a regression discontinuity across birth cohorts and linked employer–employee data from Portugal, I find that the reform raised university completion by 2.1 percentage points (8 percent) and years of university by 7 percent, with larger gains for women. Students moved towards both kinds of reformed field and away from fields the reform left unchanged. Where degrees were split, more students completed both parts. At age 30, hourly wages are 3 percent higher, and about two-thirds of the gain reflects work in higher-paying firms. Italian data show similar gains. A model in which students learn about their match while enrolled reproduces the education responses. Splitting long degrees reduced what students stood to lose if the field proved a poor match, since they could now leave early with a bachelor's. This drew in students, some of whom went on to complete the master's.

# 1 Introduction

University is a costly investment made under uncertainty. It is costly because students pay tuition and forgo earnings while enrolled. It is uncertain because students enrol with imperfect information about how likely they are to graduate and learn about their prospects only while studying. This uncertainty matters because much of the return depends on completing the degree.<sup>1</sup> Costs therefore accrue continuously, whereas returns are lumpy. Together, cost and uncertainty may deter some students from enrolling, including some who would have graduated. Degree length shapes both. Longer degrees mean more years of costs and a longer period over which the payoff remains uncertain. This paper asks how degree length shapes students' educational choices and subsequent earnings.

The Bologna Process provides a large-scale natural experiment to study this question. Signed by 29 countries in 1999, it now covers the 49 members of the European Higher Education Area. It replaced heterogeneous national degree systems with a common structure of bachelor's, master's and doctoral degrees, moving much of Continental Europe towards the shorter, sequential model of Anglo-Saxon countries. It did so in two ways. First, when a field was already split into a bachelor's and master's degrees, it shortened the bachelor's degree from four years to three. Second, when a field was organised as a single long degree of five years, it introduced a bachelor's degree after three years where none had existed before. The first mainly lowered the cost of a degree, but possibly also the learning embodied in it. The second mainly reduced the downside of uncertainty, since students who discovered that the field was a poor match could now leave after three years with a qualification. Advocates argued that shorter degrees would widen access, pointing to cross-country attainment gaps: in 2004, 39 percent of US adults aged 25–64 held a tertiary qualification, compared with 23 percent of European adults in the same age range (OECD, 2006). Critics worried that employers would not value the new degrees.

Whether advocates or critics were right is unclear a priori. On the extensive margin, lower cost and a smaller downside from uncertainty may draw in students who were previously deterred from enrolling, and some of them may discover that they are well matched and continue to a master's. On the intensive margin, students who would have enrolled anyway may stop after three years instead of four or five. Whether

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<sup>1</sup>Add citations and evidence from PIAAC data.

average schooling rises or falls depends on which response dominates. The effects on earnings are equally ambiguous. For students who stop earlier, the reform brings work experience forward but may mean less training. For students drawn into university, it delays entry into work but may raise later earnings. And because the reform changed fields in different ways, it may also shift which fields students choose, and hence the occupations they can access.

Comparing cohorts making their university choices just before and just after the reform, I ask four questions. First, did the reform raise educational attainment and earnings, on net? Second, how did students' choices of field change, and did these changes carry into occupations? Third, did the new three-year degrees lead students to stop after three years, or did some continue to longer qualifications? Fourth, what share of any wage effect is accounted for by workers sorting into higher-paying firms? I answer these questions with a regression discontinuity (RD) design using birth cohorts as the running variable, in two settings where the rollout was fast enough to create a clear break in exposure: Portugal, where the reform was implemented in 2006–07, and Italy, where it took effect in 2000–01.

Portugal is the main setting because of unusually rich linked employer–employee data. *Quadros de Pessoal* covers the Portuguese private sector and lets me observe workers' education, wages, occupations, sectors, job titles and employers, and estimate firm wage premia. Because it covers only private-sector employees, and the reform might affect who works in the private sector, I validate the results with the Portuguese Labour Force Survey and the Education and Training Survey, which provide representative samples of the resident population. The Italian Labour Force Survey data then allow me to test the external validity of my findings. Because the Italian reform took effect several years earlier and Italian students enter university at a different age,<sup>2</sup> this comparison also helps address the concern that the Portuguese discontinuity captures contemporaneous macroeconomic conditions rather than the reform itself.

The analysis yields four main findings. First, the reform raised educational attainment and wages, and both effects are larger for women. In Portugal, university completion rose by 2.1 percentage points, or 8 percent of the pre-reform mean. Average years of university rose by about 0.14 years, or 7 percent. At age 30, post-reform cohorts earn 3 percent more per hour: 3.5 percent for women and 2.3 percent for men.

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<sup>2</sup>19 in Italy, 18 in Portugal.

They enter work later, so they experience lower cumulative earnings in their 20s. However, by age 30, the early shortfall has been compensated for. Italian data show similar gains, concentrated among women. Second, the reform changed what students study. Students moved towards social sciences, economics and humanities, where the first degree became shorter, and towards engineering and other STEM fields, where the reform introduced an earlier credential. They moved away from fields left unchanged, most notably nursing in Portugal, and fewer subsequently worked as nurses. Fewer students graduated in nursing, and fewer worked as nurses. Third, students did not simply stop at the new three-year degrees. Engineering is the clearest case: employment rose in engineering jobs requiring the full five-year cycle, implying that more students completed the full qualification. Finally, the reform changed where graduates work. An AKM decomposition (Abowd et al., 1999) shows that about two-thirds of the wage gain at 30 is accounted for by workers entering firms with higher estimated wage premia. A separate decomposition by employer sector shows that IT and engineering, business services and finance together account for almost two-thirds of the total wage gain. These sectors hire intensively from the fields towards which students moved, suggesting a link between the change in educational choices and the change in employers.

The education responses map onto the two features of the investment: cost and uncertainty. Where the first degree was cut from four years to three, the reform lowered its cost, and students moved towards these fields. Engineering is different. The full programme still took five years, but students could now earn a degree after three. This did not reduce uncertainty itself; it reduced its downside. Students enter uncertain about their match and learn while enrolled. Before the reform, those who chose not to complete the full five years left without an engineering degree; afterwards, they could leave after three years with a qualification. This made engineering less costly to try, drawing in students, some of whom discovered that they were well matched and completed the full five years.

[Work in progress] To support this interpretation, I build a model of degree choice. Students start with a prior belief about how well they match a field, learn about that match while enrolled, and decide whether to continue. I estimate the model to match the education responses to the reform, including completion, choices across fields and progression to longer qualifications. The estimated model reproduces their broad pattern: more students enrol, they move from nursing towards STEM and the fields whose degrees were shortened,

and more complete five years in fields such as engineering. Preliminary counterfactuals indicate that both uncertainty and learning matter. Although wages are not targeted in estimation, the model predicts a wage gain close to the one in the data.

Existing research on the Bologna Process has focused mainly on school leavers' enrolment decisions and on dropout or on-time graduation conditional on enrolment (Cappellari and Lucifora, 2009; Cardoso et al., 2008; Horstschräer and Sprietsma, 2015; Bondonio and Berton, 2018). Evidence on earnings remains limited, mainly descriptive (Kroher et al., 2021, for a review), and generally conditions on university completion (Bosio and Leonardi, 2010; Glauser et al., 2019; Suleman and Figueiredo, 2020; Avdeev, 2025). This paper makes four contributions. First, it provides, to my knowledge, the first unified causal account of how the Bologna reform changed both education and labour-market outcomes. By estimating effects across cohorts rather than only among university entrants or graduates, I capture the combined responses graduation, years of study, field choice and earnings. The results show that the reform did more than change the credentials held by existing students: it changed who acquired a university education, what they studied and where they subsequently worked. Second, the paper changes how the reform should be understood. Existing work has mostly treated Bologna as a reduction in degree length, and hence in the cost of graduating. I show that moving the first credential forward also reduced the downside of uncertainty. The new three-year degree gave students who discovered that they were poorly matched a qualification with which to leave, thereby raising the value of trying university. This connects the paper to models of schooling as experimentation, in which students learn about their match while enrolled (Manski, 1989; Stinebrickner and Stinebrickner, 2012; Stange, 2012), and to stepping-stone models in which an earlier credential creates option value (Trachter, 2015). Existing work typically quantifies the value of options already present in the education system. I instead use a policy-induced change that created a new option in the data. Third, the paper broadens the empirical literature on stepping stones in higher education. Existing evidence comes mainly from US community colleges, where the intermediate credential is provided by a separate institution (Rouse, 1995; Jepsen et al., 2014; Stevens et al., 2019; Mountjoy, 2022). Bologna instead inserted an intermediate credential within the university programme itself. This setting shows that a stepping stone need not divert students from longer qualifications: by making the initial investment less costly under uncertainty, it can draw in

new students and increase progression to the full degree. Finally, I overcome an important limitation of the US community-college literature, which relies largely on unemployment-insurance records that measure earnings but provide little information about the jobs and employers through which those gains arise. The Portuguese linked employer–employee data let me distinguish hourly wages from employment and hours, and trace the wage effect through occupations, employment sectors and firm wage premia. In this respect, the paper builds on [Portugal et al. \(2024\)](#), who use the same data to decompose the returns to compulsory schooling in Portugal. I extend this approach to higher education and show what part of the wage effect can be attributable to firms.

The rest of the paper is structured as follows: section 2 describes the institutional setting; section 3 presents the data sources and explains sample construction; section 4 explains the empirical strategy; section 5 presents the results while section 6 outlines the model [work in progress].

## 2 Institutional Setting

The Bologna Declaration of 1999 committed European countries to a common structure of higher education: a first cycle of at least three years, followed by a master’s. Countries adopted the reform at different times and in different forms ([Cardoso et al., 2008](#)). Portugal and Italy both chose a three-year first degree followed by a two-year master’s, and both countries gave a short time span of around two years to universities to fully implement the reform.

**Timeline.** Both roll-outs were fast (Figure 1). In Portugal, Decree-Law 74/2006 let programmes convert from 2006/07 and required them to by 2008/09; 43 percent converted in the first year and about 90 percent by 2007/08 ([Cardoso et al., 2008](#)), so the 2007 applicants are the first cohort to face an almost fully reformed system. In Italy, the Ministry of Education released guidelines on how to implement the Ministerial Decree 509/1999 in August 2000. Universities whose pre-scheduled faculty meetings took place in September 2000 could implement the reform from the academic year 2000-01. In practice, most universities adapted the new regime in 2001/02 ([Cappellari and Lucifora, 2009](#); [Bondonio and Berton, 2018](#)).

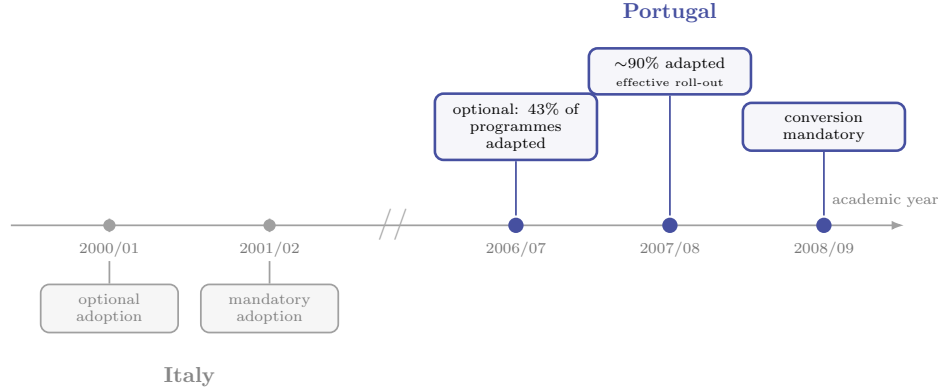


Figure 1: Implementation of the reform in Portugal and Italy

Notes: Sources: Decree-Law 74/2006; Ministerial Decree 509/1999.

**Restructuring by field.** The reform shortened the time to a first credential in two different ways (Figure 2). First, where a field was organised as a single long degree, the reform introduced a new intermediate degree after three years where none had existed before. This was the case in every reformed field in Italy, where the four- to six-year *laurea* had no degree inside it and no master's above it; it became a three-year *laurea* followed by a two-year *laurea specialistica*. In Portugal it was the case in university engineering and computing, which lasted five years; this was replaced either by a three-year bachelor's and two year master's, or an integrated master's that by law awards the first degree after three years (Decree-Law 74/2006, art. 19). The path to the profession still takes five years, but a student who leaves after three now leaves with a degree instead of nothing. Second, where a field was already split into a first degree and a master's, the reform shortened the first degree. This happened only in Portugal. The *licenciatura* in economics, management, the sciences, the social sciences and the humanities lasted four years (five at some institutions) and could be followed by a two-year master's, which about one graduate in ten took; the reform cut it to three years. Economics at the University of Porto illustrates what was removed from an economics *licenciatura*. Between the four-year plan of 2004/05 and the three-year plan of 2008, the degree lost Microeconomics III, Mathematics III, Econometrics II, and four of its twelve applied field courses (international monetary economics, innovation, regional economics and economic policy), while several of the remaining field courses shrank from

six to four credits and the number of management and finance courses rose from six to nine.<sup>3</sup> The first cycle became the old degree minus most of its final year, which moved into the master's. Some fields maintained the same structure after the reform. Medicine and law kept long degrees of five to six years in both countries. Nursing stayed four years in Portugal; in Italy it was already a three-year university diploma, but became a recognised university degree followed by a master's.

|                                                  | Portugal |      | Italy |       |                                                                   |
|--------------------------------------------------|----------|------|-------|-------|-------------------------------------------------------------------|
|                                                  | pre      | post | pre   | post  |                                                                   |
| Engineering & computing                          | 5        | 3+2  | 5     | 3+2   | new intermediate credential                                       |
| Economics, sciences, humanities, social sciences | 4(+2)    | 3+2  | 4     | 3+2   | PT: shorter first-cycle degree<br>IT: new intermediate credential |
| Nursing                                          | 4        | 4    | 3     | 3(+2) | PT: no change<br>IT: diploma renamed laurea                       |
| Medicine, law                                    | 5–6      | 5–6  | 5–6   | 5–6   | no change                                                         |

Figure 2: Years to the first university credential before and after the reform, by field

Notes: Statutory years to the first degree. (+2): pre-existing master's. Blue: new intermediate credential; orange: shorter first-cycle degree. Polytechnic and short-diploma programmes not shown.

**Enrolment: demand, supply and age at entry** The increase in enrolment was driven by demand, not by supply of places. In Portugal each institution proposes the number of places per programme and candidates are allocated through a national contest. Places grew by 5 percent in 2007, while candidates rose by 27 percent (Figure 3a); the extra students filled places that already existed, since in 2006 one place in four had gone unfilled. In Italy most programmes had no cap on entry, so enrolment was not rationed by places at all; first-time entrants rose by 12 percent in 2001/02 and stayed about 18 percent above their 1998–2000 level. The reforms did not change the age at which students enter (Figure 3b).

<sup>3</sup>Deliberations 396/2004 and 397/2004 (*Diário da República*, 26 March 2004) and the faculty's official Bologna study plan (2008). The removed material reappears as options or in the new master's in economics. Cardoso et al. (2008) describe the new curricula more generally as compressed versions of the old ones.

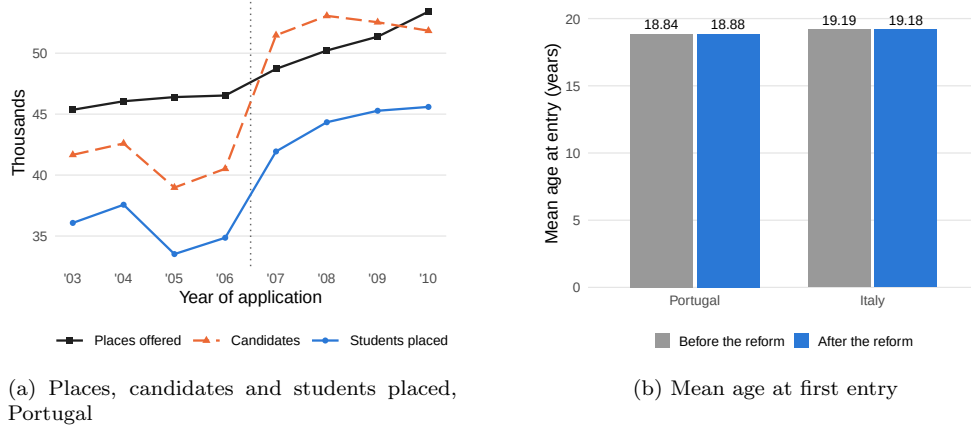


Figure 3: Enrolment: places versus demand, and age at entry

Notes: (a) Data from DGES. (b) Data from Eurostat.

### 3 Data

#### 3.1 Data sources

I combine one administrative dataset and three surveys. *Quadros de Pessoal* is the backbone of the Portuguese analysis: it follows every private-sector employee over time and records education, field of study, occupation, job title, pay and employer for the full cohorts on both sides of the reform. Two Portuguese surveys fill the gaps it leaves. The Labour Force Survey covers the people *Quadros de Pessoal* never sees—public-sector workers, the self-employed and those out of work—and the Education and Training Survey records who entered higher education and who left without a degree. The Italian Labour Force Survey supplies data for the Italian analysis, which I use to test whether the Portuguese patterns travel elsewhere.

**Quadros de Pessoal.** *Quadros de Pessoal* (QP) is a linked employer–employee data set collected every year by the Portuguese Ministry of Labour from all private-sector establishments with at least one employee. Reporting is compulsory, and each establishment reports its workforce as of October. QP therefore covers the full population of workers and firms in manufacturing and private services; it excludes civil servants, the self-employed and domestic workers. For each worker I observe sex, birth date, the highest completed level of education and its field, a four-digit occupation code, the job title within the applicable collective agreement (*categoria profissional*), the worker’s hierarchical level within the firm (*nível de qualificação*), normal monthly hours, and monthly base pay, regular supplements and overtime pay. Workers and firms

carry anonymised identifiers, so I can follow a worker across employers and years. I use the waves from 2000 to 2023; no worker file exists for 2001.

The QP education variable measures well whether a worker holds a university degree, but it cannot reliably tell one level of degree from another. Before 2006 the education classification has no separate code for the master’s degree, which is pooled with the *licenciatura*; meaning I can distinguish master’s from first-cycle degrees only from 2006 onwards. Splits of the degree effect by level and length in QP are therefore less reliable than the effect on holding any degree. For years of university, I rely on a proxy calculated as the age an individual first appears in QP minus 18 for those who attended university, and 0 otherwise. I validate this imperfect measure using the Portuguese LFS, which record the age at which a person left education, and on the Italian LFS, which additionally codes the length of the degree directly.

**Portuguese Labour Force Survey.** The *Inquérito ao Emprego* is the quarterly household survey of Statistics Portugal (INE) and is representative of the resident population. Unlike QP it covers every employment status, so it measures educational attainment without conditioning on private-sector employment nor on any type of employment. It records the highest completed level of education and its field, the age at which the respondent left education, employment status and monthly pay, with education codes close to those in QP. I use the quarterly files from 1998 to 2024.

**Education and Training Survey.** The *Inquérito à Educação e Formação de Adultos* (IEFA) is INE’s module of the European Adult Education Survey. I use its four waves, 2007, 2011, 2016 and 2022. It is the only Portuguese source that records whether a respondent ever enrolled in higher education and left without a degree; it also records the level and field of the highest degree and, from 2016, whether that degree was a post-Bologna degree. The samples are small—about 3,600 women and 3,500 men in the cohort window—so I use IEFA for enrolment and dropout rather than for precise effects on attainment.

**Italian Labour Force Survey.** The *Rilevazione sulle Forze di Lavoro* is the quarterly household survey of the Italian statistics office (ISTAT). I use the public-use quarterly files from 2009, the first year that reports year of birth, to 2020. The survey records the highest completed level of education and separates the two-cycle degrees created by the reform—the three-year *laurea* and the two-year *laurea specialistica*—from

old-regime and single-cycle degrees. It also records the degree subject, the age at which the respondent left education, employment status, occupation and net monthly pay in the main job for employees, top-coded at 3,000 euros. I add ministry counts of students enrolled in two-cycle programmes by university and year, which identify the regions whose universities adopted the reform in 2000/01.

### 3.2 Sample construction

The design compares people who reached the age of university entry just before and just after the reform, so the unit of comparison is the entry cohort. I build every sample in two steps. I first assign each person to the year in which they faced the decision to enrol. Then, I measure that year relative to the year the reform reached them.

**Portugal.** Portuguese students apply to university at the end of secondary school, usually at 18, so I assign each person to the calendar year in which they turned 18,  $c_i = \text{year of birth} + 18$ . The reform reached almost every programme within one year: 43 percent of programmes converted in 2006/07 and about 90 percent by 2007/08 (Section 2). The cohort that turned 18 in 2007, born in 1989, is therefore the first to apply to an effectively fully reformed system, and the running variable is  $Z_i = c_i - 2007$ . The 2006 cohort applied when fewer than half of the programmes had converted. It is only partly exposed, so I exclude it rather than assign it to either side. I keep the six cohorts on each side of this gap, those turning 18 in 2000–2005 and in 2007–2012 (born 1982–1987 and 1989–1994). The same twelve cohorts define every Portuguese sample.

**Italy.** In Italy the reform did not reach every student in the same year, so the cutoff depends on where a person studied. To date it, I use ministry counts of students enrolled in two-cycle and in old-regime programmes at each university from 2000/01, aggregated to the region in which the university is located. In 2000/01 only four regions had a non-negligible share of students in two-cycle programmes: Piemonte, Valle d’Aosta, Friuli-Venezia Giulia and Sardinia. I call these the early adopters, which converted in 2000/01, and the rest of the country the late adopters, which converted in 2001/02. The Labour Force Survey does not record where a respondent studied, so I assign each respondent to the adoption year of their region of residence. This is a close proxy, because regional mobility in Italy is low: 84 percent of graduates live in the

region where they studied (Bondonio and Berton, 2018). Italian students usually enter university at 19, so with  $c_i = \text{year of birth} + 19$  the running variable is

$$Z_i = \begin{cases} c_i - 2000 & \text{if } i \text{ lives in an early-adopting region,} \\ c_i - 2001 & \text{if } i \text{ lives in a late-adopting region.} \end{cases} \quad (1)$$

The survey data confirm the classification: the share of respondents with a two-cycle degree jumps one cohort earlier in the early-adopting regions than in the rest of the country (Appendix Figure 10).

The resulting samples contain 12.6 million QP worker-years from 1.8 million workers, 310,355 respondents in the Portuguese Labour Force Survey, about 7,100 IEFA respondents and 269,026 respondents in the Italian survey. Appendix B.1 defines the education and labour-market outcomes.

### 3.3 Summary statistics

Table 7 reports raw means for the cohorts before and after the reform in each source. The gap between columns mixes the reform, the cohort trend and differences in the age and year at which each cohort is observed. **[TK: To add]**

## 4 Empirical Strategy

I compare birth cohorts that reached the age of university entry just before and just after the reform. Because the reform reached almost every university programme within a single year (Section 2), whether a student faced the old or the new degree structure depended on the year they were born, not on anything they chose. This section sets out the regression discontinuity design. Appendix C reviews the evidence that supports the design, and describes the decomposition I use to split the wage effect into a firm and a worker component.

Let  $c_i$  be the calendar year in which person  $i$  turns 18, the usual age of application in Portugal. The running variable is  $Z_i = c_i - 2007$ , and the treatment indicator is  $D_i = \mathbf{1}[Z_i \geq 0]$ : the cohort turning 18 in 2007 is the first to apply to an almost fully reformed system. The 2006 cohort applied when 43 percent of programmes had converted and is only partly exposed, so I drop it. I keep six cohorts on each side, those turning 18 in 2000–2005 and in 2007–2012.

For an outcome  $Y_{it}$  of person  $i$  observed in year  $t$  at age  $a_{it}$ , I estimate

$$Y_{it} = \alpha_0 + \alpha_1 D_i + \alpha_2 Z_i + \alpha_3 D_i Z_i + g(a_{it}) + \varepsilon_{it}, \quad (2)$$

where  $g(\cdot)$  is a quadratic in age. The coefficient of interest is  $\alpha_1$ , the jump in the outcome at the cutoff. The terms in  $Z_i$  allow a separate linear cohort trend on each side, so  $\alpha_1$  compares the first reformed cohorts with what the trend of the pre-reform cohorts predicts for them. Outcomes are the education measures and labour-market outcomes defined in Appendix B.1.

The design requires that, without the reform, outcomes would have continued smoothly across the 2007 cutoff. Four pieces of evidence support it. First, people cannot choose the year they were born, and the number of workers per cohort in *Quadros de Pessoal* shows no break at the cutoff (Figure 11, Appendix C.2). Second, students who turned 18 before 2007 did not wait to enter the new degrees. If they had, more of them would have worked between leaving secondary school and graduating from university; the discontinuity in that share is 0.1 percentage points on a base of 10.5 percent, and it is close to zero for both women and men (Table 8, Appendix C.2). Third, the mean age at first entry did not change (Figure 3b), which supports assigning exposure by the year a cohort reaches the usual age of entry. Fourth, *Quadros de Pessoal* sees only private-sector employees, so a reform that changed who works in the private sector would show up as a change in the sample. I validate that every education result replicates in the Labour Force Survey, which covers the whole population.

For Italy I estimate the same equation. However, in Italy I do not use a donut setting because the running variable is the year a person turns 19 minus the year universities in their region of residence adopted the reform, 2000 or 2001 (Section 3.2).

## 5 Results

This section reports the reform’s effects in the order in which a student meets them. I start with education: whether more students completed a university degree, how long they studied, which fields they chose, and whether they stopped at the new three-year degree or continued. I then turn to the labour market: hourly wages at 30, how much of that gain comes from better-paying firms and from pay above the job-title floor,

and cumulative earnings over the life cycle. Throughout, I report effects separately for women and men, since the reform affected them differently at almost every margin. Each estimate is the coefficient  $\alpha_1$  in equation (2), and “baseline” is the mean of the outcome in the last cohort before the reform.

**More graduates and more years of university** The reform raised university completion. Figure 4 plots the share of each cohort whose highest completed level is a university degree, together with average years of university in the population, counting zero for those who never graduated. Both series jump at the first reformed cohort, and the jump persists in the cohorts that follow. Among women the share with a degree rises by 2.9 percentage points, from a baseline of 34 percent, or 8.5 percent; among men it rises by 1.4 percentage points from 20 percent, or 7 percent. The effect on women is twice as large in levels and significantly different from the effect on men.

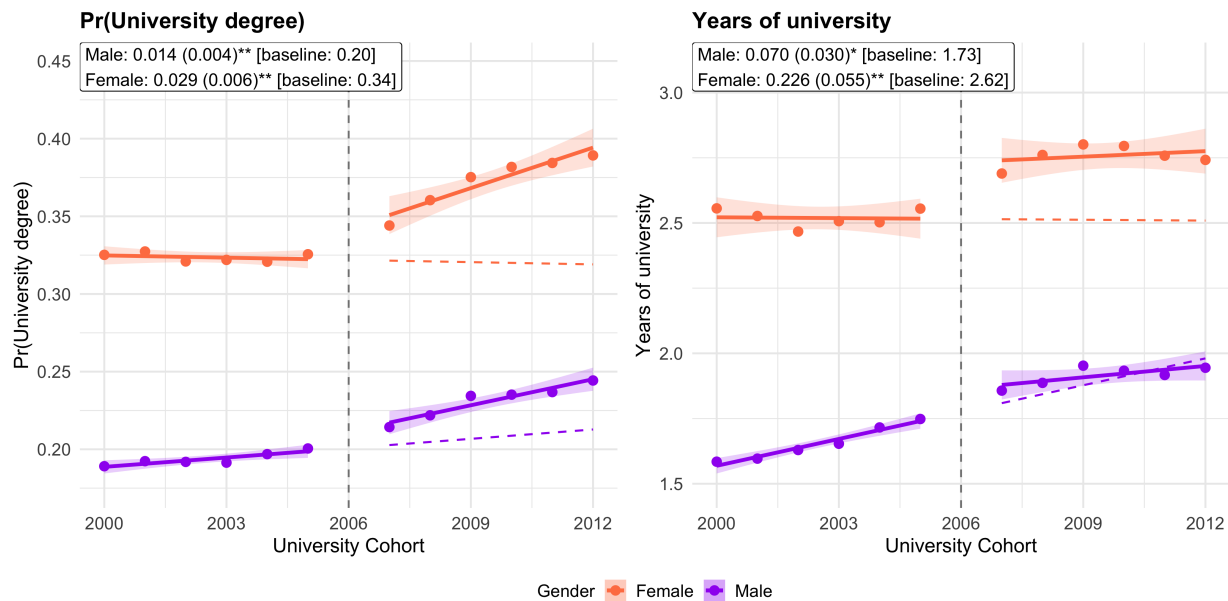


Figure 4: University completion and years of university, by cohort

*Notes: Quadros de Pessoal.* Dots are cohort means; solid lines are linear fits on each side of the cutoff with 95 percent confidence bands; dashed lines extend the pre-reform fit. Cohorts are defined by the year a person turned 18; the 2006 cohort is excluded. Left panel: share whose highest completed level is a university degree. Right panel: years of university, zero for non-graduates. Boxes report the RD estimate, its standard error and the pre-reform baseline.

Because the reform shortened the first degree, more graduates need not mean more schooling. Students who would have completed a four- or five-year degree could instead stop after three, and this loss could offset the gain from new graduates. It does not. Average years of university in the population rise by about

0.2 years for women and 0.07 to 0.09 years for men (Table 1), roughly 10 percent and 4 to 7 percent of the baseline. The new graduates therefore add more schooling than any early stoppers remove: total university education in each cohort rises, even though the shortest route to a degree became one or two years shorter.

Table 1 shows that the result does not depend on *Quadros de Pessoal*. Because QP observes only private-sector employees, a reform that changed who works in the private sector could produce a jump in attainment within QP without any change in the population. The Labour Force Survey, which covers the whole resident population, gives a larger effect on women’s completion—3.7 percentage points on a lower baseline of 23 percent—and a similar effect on men’s, 1.2 percentage points. The effects on years of university are also close across the two sources. The Education and Training Survey points in the same direction for both outcomes, but with about 3,500 respondents of each sex its estimates are too imprecise to distinguish from zero.

Table 1: Effect of the reform on university completion and years of university, three data sources

|                              | Highest education is a university degree |                     |                  | Years of university |                     |                  |
|------------------------------|------------------------------------------|---------------------|------------------|---------------------|---------------------|------------------|
|                              | QP                                       | LFS                 | IEFA             | QP                  | LFS                 | IEFA             |
| Women                        | 0.029***<br>(0.006)                      | 0.037***<br>(0.004) | 0.056<br>(0.041) | 0.205***<br>(0.006) | 0.185***<br>(0.024) | 0.281<br>(0.263) |
| Baseline                     | 0.340                                    | 0.229               | 0.287            | 1.919               | 1.938               | 1.811            |
| Men                          | 0.014***<br>(0.004)                      | 0.012***<br>(0.003) | 0.049<br>(0.053) | 0.092***<br>(0.005) | 0.070***<br>(0.020) | 0.147<br>(0.357) |
| Baseline                     | 0.200                                    | 0.133               | 0.121            | 1.274               | 1.608               | 1.730            |
| <i>p</i> -value, women = men | 0.000                                    | 0.000               | 0.921            | 0.000               | 0.000               | 0.762            |

*Notes:* RD estimates at the 2007 cohort, excluding the 2006 cohort, with separate linear trends in the running variable on each side and age controls. Standard errors in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Baseline: mean in the last pre-reform cohort. Observations (women / men): QP 6,153,392 / 6,797,302 worker-years; LFS 187,114 / 191,608 person-quarters; IEFA 3,648 / 3,532 persons (weighted). Years of university are zero for non-graduates; in QP they are proxied by the age at which a worker is first observed with their highest degree, minus 18 (Appendix B.1).

The reform could also have raised completion by lowering dropout among students who would have enrolled anyway. IEFA is the only source that records leaving university without a degree. The RD estimate of the change in the dropout rate is  $X$  percentage points. **[TK: Add this!]**

**Students moved towards the fields the reform changed.** The reform did not treat all fields alike, and students responded to the differences. I group fields by what the reform did to the time to a first credential (Figure 2): fields where a five-year degree gained a three-year exit (“5 to 3+2”, mostly engineering and the sciences); fields where a four-year first degree became three years (“4+2 to 3+2”, economics and management, the social sciences, the humanities and the arts); and fields with no usable early exit (“no change”, health,

including nursing, and law). The outcome is the share of the whole cohort holding a degree in each group, so it combines students drawn into university with students who would have enrolled anyway but chose a different field.

Figure 5 shows three results. First, degrees in fields that gained a three-year exit rise by 2.1 percentage points for women, about a third of their baseline of 6 percent, and by 1.3 percentage points for men, about 12 percent of their baseline of 10 percent. Second, degrees in fields where the first degree became shorter rise by 2.6 percentage points for women, a fifth of their baseline, and by 0.6 percentage points for men. Third, degrees in the unchanged fields fall, by 2.6 percentage points for women and 0.6 percentage points for men, in both cases more than a quarter of the baseline. For women, most of this fall is nursing, which stayed a four-year degree in Portugal and so became long relative to the alternatives. The rise in the two reformed groups is larger than the overall rise in graduates, so the reform did more than add students: it also moved students who would have graduated anyway out of the unchanged fields.

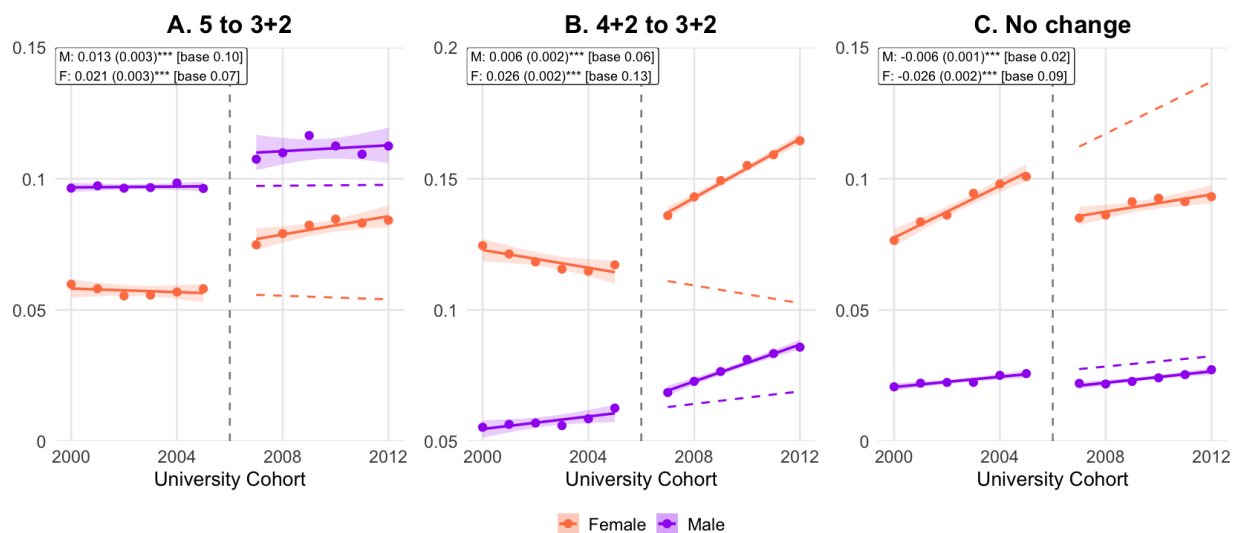


Figure 5: University degrees by what the reform did to the field, by cohort

Notes: *Quadros de Pessoal*. Share of the cohort whose highest completed level is a university degree in each group of fields. Panel A: fields where a five-year degree became three plus two (STEM). Panel B: fields where a four-year first degree became three years. Panel C: fields with no usable early exit (health and law). See the notes to Figure 4.

The shift across fields shows up in the jobs people hold. With occupation as the outcome, the share of women working in nursing occupations falls by about 3 percentage points, while the shares working in STEM and in economics and business occupations rise by 1 to 1.5 percentage points each; the changes for men are

smaller and have the same sign (Appendix Table [TK: ref]).

The same pattern appears in the Labour Force Survey, which cannot separate law from the social sciences and business: degrees rise in the reformed fields and fall in health and law, with larger changes for women (Appendix Table [TK: ref]).

**Students did not stop at three years** The critics’ worry was that students would take the new exit and leave with a three-year degree that employers value less. Engineering provides a direct test. In Portugal the title of engineer, and the acts reserved for it, require the full five-year cycle, so an engineering occupation reserved for full-cycle graduates can only be reached by students who continued past the third year (See A.2. If the reform had mainly diverted engineering students to the three-year exit, employment in these occupations would not rise.

It rises. Figure 6 shows that engineering degrees rise by 1.3 percentage points for women and 0.9 percentage points for men. Employment in engineering occupations rises by 0.9 and 0.5 percentage points, and employment in the engineering occupations that require the five-year degree rises by 0.6 percentage points for both sexes, roughly half of the women’s baseline and a fifth of the men’s. Pooling women and men, the reform raised engineering degrees by 1.1 percentage points on a base of 4.2 percent, and employment in five-year engineering occupations by 0.6 percentage points on a base of 2.3 percent. Before the reform, 56 percent of engineering graduates worked in such occupations. If the new graduates took up these jobs at the same rate, the 0.6 point rise implies that about 1.1 percentage points more of each cohort completed the full five years—as many as the rise in engineering degrees itself. The reform brought students into engineering, and a large share of them went on to the end.

Outside STEM the reform shortened the first degree rather than adding an exit, and the question is whether the new graduates also continued to a master’s. Of the additional non-STEM graduates, 17 percent hold a master’s degree, almost the same as the 18 percent of pre-reform non-STEM graduates.<sup>4</sup> The marginal graduate in these fields is about as likely to continue as the average graduate before the reform.

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<sup>4</sup>Both shares come from *Quadros de Pessoal*, which codes the master’s degree separately only from 2006 and uses a single *licenciatura* code for degrees of different lengths. The split between first-cycle and master’s degrees is therefore less reliable than the effect on holding any degree (Section 3).

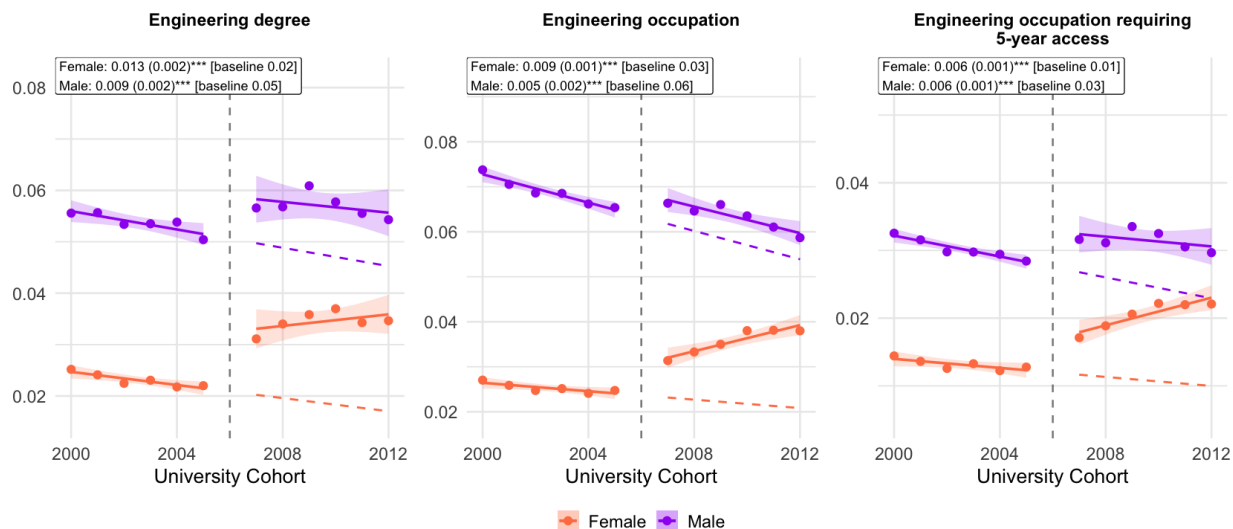


Figure 6: Engineering degrees and engineering occupations, by cohort

Notes: *Quadros de Pessoal*. Left: share of the cohort holding an engineering degree. Middle: share working in an engineering occupation. Right: share working in an engineering occupation that requires the full five-year cycle. See the notes to Figure 4.

**Italy validates the key findings** Italy reformed its universities along the same lines six years earlier, and its students enter at a different age, so a Portuguese jump caused by something else that happened to the 2007 cohort would not reappear there. Figure 7 puts the two countries side by side, with Italian fields grouped by the same rule. In Italy, degrees in the fields that gained a three-year exit rise by 1.2 percentage points for women and 0.7 percentage points for men; degrees in fields whose four-year degree became three plus two rise by 0.9 percentage points for women and do not change for men. The share of each cohort without a university degree falls by 1.8 percentage points for women and 0.8 percentage points for men, compared with 2.2 and 1.3 percentage points in Portugal. The Italian field effects are about half the Portuguese ones, and in both countries they are larger for women.

The Italian survey separates degrees of four or more years from shorter ones, so it can check directly whether students stopped at three years. Figure 8 shows the answer for women. Engineering degrees rise by 0.4 percentage points, and engineering degrees of four or more years by 0.3 percentage points, so most of the new engineering graduates completed a long degree. Healthcare offers a contrast with Portugal. In Italy nursing was already a three-year diploma, and the reform turned it into a first degree with a master's on top. Nursing therefore did not become long relative to other fields, as it did in Portugal; instead it gained

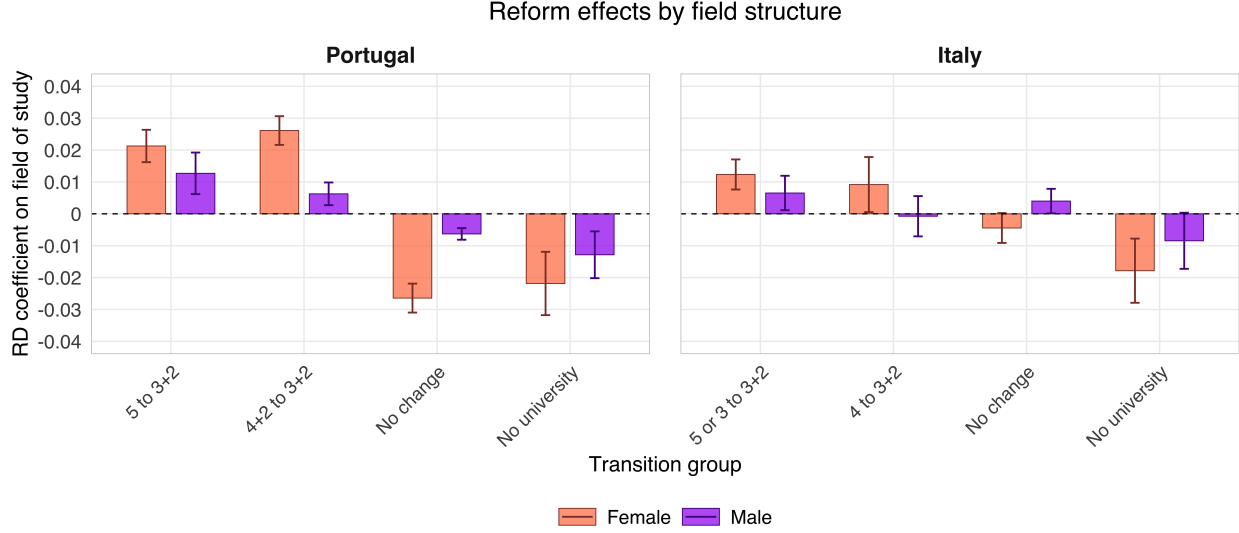


Figure 7: Effects on degrees by what the reform did to the field, Portugal and Italy

*Notes:* RD estimates with 95 percent confidence intervals. Portugal: *Quadros de Pessoal*, cohorts defined by the year a person turned 18. Italy: Labour Force Survey, citizens aged 30–36, cohorts defined by the year a person turned 19 relative to the regional adoption year. In Italy, the first group also includes the three-year *diploma universitario*, which became a first degree.

a route to further study. Healthcare degrees among Italian women rise by 0.8 percentage points, and long healthcare degrees by 0.4 percentage points. In both countries, then, students moved towards the fields where the reform added a credential and away from those where it did not.

Taken together, the education results describe a reform that brought new students into university, moved them towards the fields where an earlier credential appeared, and did not lead them to stop early. Section 6 develops a model in which students learn about their match with a field while enrolled, estimates it on these education effects, and uses it to predict the effect of the reform on wages without targeting it. The rest of this section estimates that wage effect directly.

## 5.1 Higher wages, partly from better-paying firms

Post-reform cohorts earn higher hourly wages. For the whole cohort, the log hourly wage pooled over ages and adjusted for age and year (Appendix C.1) jumps by 1.7 to 1.8 log points at the cutoff. Table 2 measures the effect at a single age, 30. At a single age cohort and calendar year move together, so year effects cannot be removed; instead I subtract from each wage the mean log wage of workers aged 40 to 45 in the same year, with their composition held at its 2005 value. At 30, post-reform cohorts earn 3.0 percent more. The effect

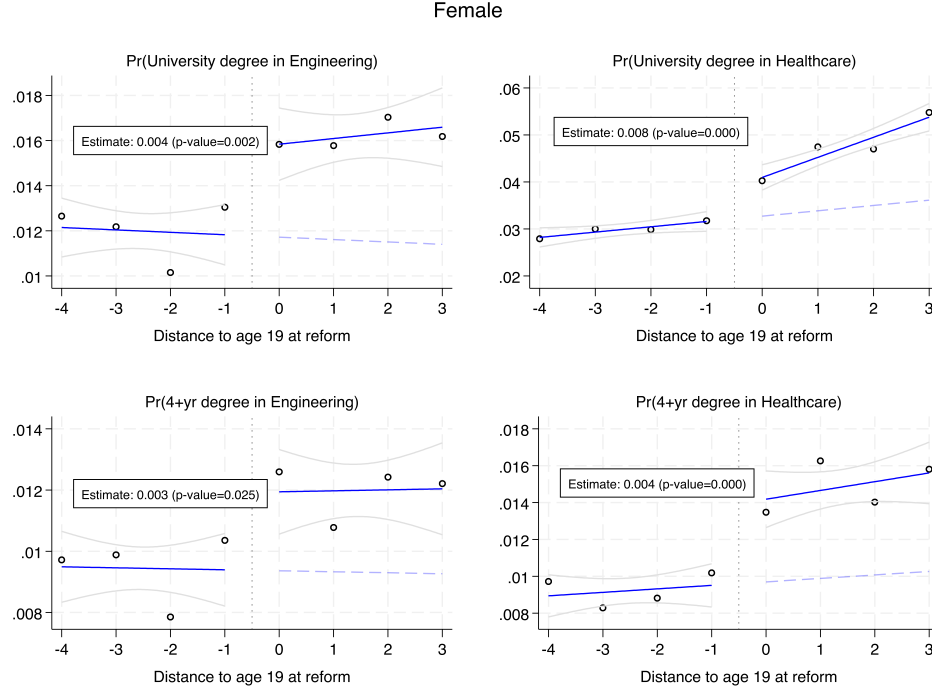


Figure 8: Engineering and healthcare degrees by length, Italian women

Notes: Italian Labour Force Survey, women aged 30–36. The running variable is the year a person turned 19 minus the adoption year in their region. Top: any university degree in the field. Bottom: a degree of four or more years (single-cycle or *laurea specialistica*) in the field.

is 3.5 percent for women and 2.3 percent for men; the larger point estimate for women matches their larger gains in completion and years of university, but the difference between them is not statistically significant.

To see how much of this gain comes from where people work, I decompose the age-adjusted wage into a worker effect, a firm effect and year effects (Abowd et al., 1999), and estimate the RD on the firm effect of each worker’s employer (Appendix C.3). Because the same decomposition applies to every worker, the RD on the wage equals the RD on the firm effect plus the RD on the rest. Column 2 shows that at 30, post-reform women work at firms that pay 2.2 percent more to all their workers, and men at firms that pay 1.6 percent more. Sorting into better-paying firms thus accounts for about two-thirds of the wage gain: 65 percent for women and 68 percent for men. The remaining third is the part of the wage that moves with the worker, such as their degree, field and occupation, and on its own it is not statistically different from zero. The firm share is similar for women and men.

Which firms are these? Because the reform changes who works where, comparing wages within a type

of firm would compare different people. Instead, for every worker I take the log wage minus its pre-reform mean if the worker is employed in a given type of firm, and zero otherwise, and estimate the same RD on it (Appendix C.4. The four types in Table 3 cover every worker, so the four RDs add up exactly to the wage effect. IT and engineering firms account for 28 percent of the effect, and business and support services firms together with banking, insurance and finance for another 35 percent.

A third split asks how the gain is paid, following the approach in [Cardoso and Portugal \(2005\)](#) explained in Appendix C.5. Most Portuguese private-sector workers are covered by collective agreements that set a minimum rate for each job title. A higher wage can therefore come from a higher floor, the rate attached to the worker’s job title, or from a larger cushion, the pay a firm adds above that rate. I measure the floor of a job title in a year as the modal hourly wage of the workers who hold it, net of the same calendar index, and split each worker’s wage into the floor and the cushion. Eighty percent of workers at 30 hold a job title with a measured floor, and on this sample the wage gain is 2.4 log points (Table 4). The cushion accounts for most of it, 1.9 log points, against 0.6 for the floor, which is not distinguishable from zero. For women the cushion gain is 2.2 log points and precisely estimated; for men it is 1.4 log points and is not.

Table 2: Hourly wages at age 30: firm pay premium

|                              | Log wage<br>(1)    | Firm effect<br>(2)  | Rest<br>(3)      |
|------------------------------|--------------------|---------------------|------------------|
| All                          | 0.030**<br>(0.010) | 0.019***<br>(0.003) | 0.010<br>(0.009) |
| Women                        | 0.035**<br>(0.010) | 0.022***<br>(0.003) | 0.012<br>(0.009) |
| Men                          | 0.023**<br>(0.010) | 0.016***<br>(0.002) | 0.008<br>(0.009) |
| <i>p</i> -value, women = men | 0.43               | 0.10                | 0.72             |

*Notes:* *Quadros de Pessoal*, workers aged 30 in the largest connected set, main job, cohorts 2000–2011 (2006 excluded), observed in 2012–2023. Log wage: log real hourly wage (monthly hours and earnings winsorised at the 99th percentile within year) minus a fixed-composition index of the log wage of workers aged 40–45 in the same year (composition held at 2005; 2022 value used for 2023). Firm effect:  $\hat{\psi}_{J(i,t)}$  from equation (9); rest = log wage minus firm effect. Column 1 = 2 + 3 exactly. RD on cohort-by-sex means, linear on each side: 6 pre- and 5 post-reform cohorts. Standard errors in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Italian wages show the same gender pattern (Table 5). Among Italian women aged 30 to 36, the log net monthly wage rises by 2.1 percent at the cutoff, with no change in the probability of being employed; average years of university rise by 0.15 years, or 7 percent. Among men none of the three outcomes changes. The Italian wage effect for women is the same size as the Portuguese one, although it is measured with a different wage concept, in a different country and six years earlier.

Table 3: Hourly wages at age 30: where the effect accrues, by the type of firm people work in

|                                  | Log wage<br>(1)    | Part of the effect in        |                                         |                                          |                           |
|----------------------------------|--------------------|------------------------------|-----------------------------------------|------------------------------------------|---------------------------|
|                                  |                    | IT and<br>engineering<br>(2) | Business and<br>support services<br>(3) | Banking, insurance<br>and finance<br>(4) | All other<br>firms<br>(5) |
| All                              | 0.030**<br>(0.010) | 0.008***<br>(0.002)          | 0.007**<br>(0.002)                      | 0.003***<br>(0.001)                      | 0.011<br>(0.009)          |
| Women                            | 0.035**<br>(0.010) | 0.005***<br>(0.001)          | 0.009**<br>(0.003)                      | 0.004***<br>(0.001)                      | 0.017*<br>(0.009)         |
| Men                              | 0.023**<br>(0.010) | 0.011***<br>(0.002)          | 0.006**<br>(0.002)                      | 0.002***<br>(0.001)                      | 0.005<br>(0.009)          |
| <i>p</i> -value, women = men     | 0.43               | 0.02                         | 0.44                                    | 0.25                                     | 0.31                      |
| Share of the effect, all (%)     | 100                | 28                           | 25                                      | 10                                       | 38                        |
| Share of workers, pre-reform (%) | 100                | 4.4                          | 13.8                                    | 2.5                                      | 79.3                      |

*Notes:* *Quadros de Pessoal*, sample and log wage as in Table 2. Column 1 is the RD on the log wage. Columns 2–5 are RDs on (log wage minus its pre-reform mean) times an indicator for working in that type of firm, zero otherwise, defined for every worker; they add up exactly to column 1. No column conditions on the type of firm. A type of firm adds to the effect when more people work there at above-average pay, or when those who work there are paid more. IT and engineering: CAE 620, 631, 582, 26, 293. Business and support services: 701, 702, 692, 749 and divisions 77–82. Banking, insurance and finance: divisions 64–66. RD on cohort-by-sex means, linear on each side: 6 pre- and 5 post-reform cohorts. Standard errors in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 4: Hourly wages at age 30: job-title floor and pay above the floor

|                              | Log wage<br>(1)    | Floor<br>(2)     | Cushion<br>(3)      |
|------------------------------|--------------------|------------------|---------------------|
| All                          | 0.024**<br>(0.009) | 0.006<br>(0.011) | 0.019**<br>(0.006)  |
| Women                        | 0.034**<br>(0.010) | 0.011<br>(0.014) | 0.022***<br>(0.004) |
| Men                          | 0.014<br>(0.008)   | 0.000<br>(0.012) | 0.014<br>(0.010)    |
| <i>p</i> -value, women = men | 0.14               | 0.52             | 0.45                |

*Notes:* *Quadros de Pessoal*, sample and log wage as in Table 2, restricted to workers whose job title has a floor. Floor: modal hourly wage of the job title (*categoria profissional*) in the year, cells of at least ten worker-years, minus the same index; cushion = log wage minus floor. Column 1 = 2 + 3 exactly. RD on cohort-by-sex means, linear on each side: 6 pre- and 5 post-reform cohorts. Standard errors in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table 5: Effect of the reform on employment, wages and years of university in Italy

|                              | Employed          | Log net monthly wage | Years of university |
|------------------------------|-------------------|----------------------|---------------------|
| Women                        | 0.002<br>(0.005)  | 0.021***<br>(0.007)  | 0.147***<br>(0.040) |
| Men                          | −0.003<br>(0.005) | 0.001<br>(0.005)     | 0.041<br>(0.038)    |
| <i>p</i> -value, women = men | 0.555             | 0.022                | 0.055               |
| Baseline, women / men        | 0.602 / 0.774     | 6.912 / 7.149        | 1.973 / 1.352       |

*Notes:* Italian Labour Force Survey, citizens aged 30–36. The running variable is the year a person turned 19 minus the adoption year in their region. Net monthly pay in the main job, top-coded at 3,000 euros. Standard errors in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## 5.2 Lower earnings in the twenties, made up by the early thirties

Wages are measured only for people at work, so they miss the cost of staying longer in university. Cumulative earnings do not. I sum each person's annual earnings up to a given age, counting zero in years they are not observed in QP, and estimate the RD at each age. Figure 9 plots the estimates. Up to age 25 the reform lowers cumulative earnings, and the gap widens every year: post-reform cohorts spend more time in university and start work later. At 25 the shortfall reaches about 4,500 euros for women and 5,200 euros for men, against pre-reform cumulative earnings of about 15,000 and 17,000 euros. From then on post-reform cohorts earn more each year, and the gap closes: around age 30 for men and 31 for women. By the early thirties, the higher wages have made up the earnings lost while studying.

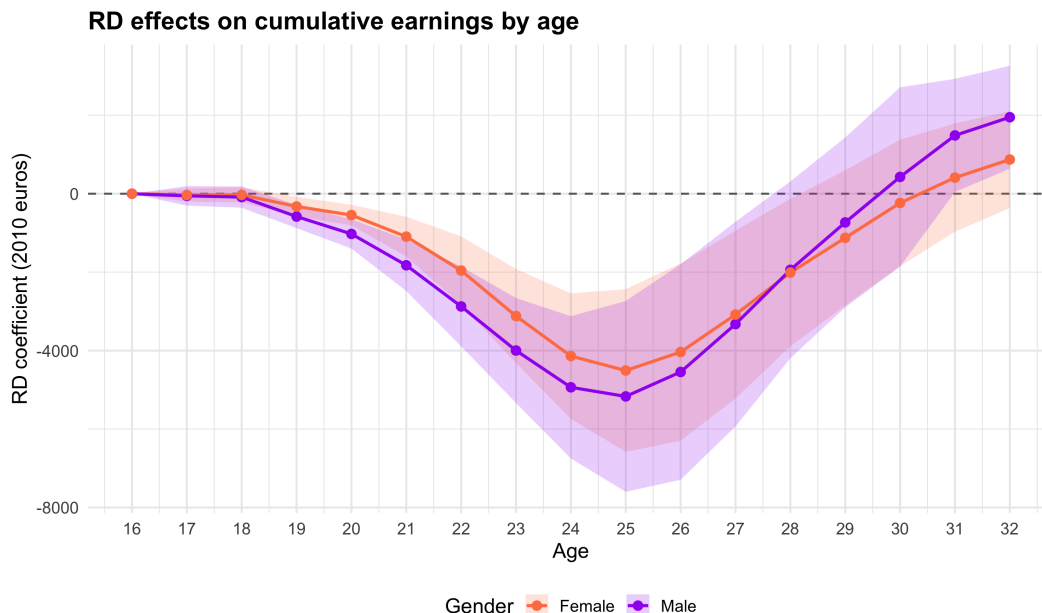


Figure 9: Effect of the reform on cumulative earnings, by age

*Notes: Quadros de Pessoal.* Each point is a separate RD estimate of cumulative annual earnings from age 16 up to the age shown, in 2010 euros (consumer price index), with years not observed in QP counted as zero, and a 95 percent confidence band. Annual earnings are October monthly earnings times twelve, summed over jobs. RD on cohort-by-sex means, linear on each side, cohorts 2000–2012 (2006 excluded). A cohort enters at an age only if it is observed at that age by 2023, so at 31 the post-reform side has four cohorts and at 32 three (2007–2009). Appendix B.1 describes the construction.

## 6 A model of learning and stopping [work in progress]

Why does moving the first credential forward change who enters university, which field they choose and how long they stay? This section writes down a simple model to address this question. A student is unsure whether she is a good match for a field, learns about it while studying, and can leave at any time with whatever credential she has reached. The reform changes one object: when the first credential arrives (Section 2). I estimate the model on the education effects alone and compare the wage effect it predicts with the one estimated in Section 5.

### 6.1 Environment

A cohort of students, each with a prior belief  $p_0 \in (0, 1)$  that she is a good match for university study, chooses between three fields,  $f \in \{\text{STEM}, \text{non-STEM}, \text{nursing}\}$ , and an outside option. Priors are distributed  $p_0 \sim \text{Beta}(1.22, 1.83)$ , with mean 0.40: the distribution of Portuguese students' own expectations of completing university in PISA 2015. A student's true match  $\theta \in \{H, L\}$  is good with probability  $p_0$  and is the same in every field; what differs across fields is how quickly study reveals it.

**Beliefs.** Each year  $t$  of study in field  $f$  produces a signal  $s_t \in \{g, b\}$ , good with probability  $\lambda_f^H$  if the match is good and  $\lambda_f^L = 1 - \lambda_f^H$  if it is bad. I call  $\pi_f = \lambda_f^H - \lambda_f^L$  the precision of the field's signals. With belief  $p$ , the probability of a good signal and the updated beliefs are

$$q_f(p) = p\lambda_f^H + (1 - p)\lambda_f^L, \quad p^g = \frac{p\lambda_f^H}{q_f(p)}, \quad p^b = \frac{p(1 - \lambda_f^H)}{1 - q_f(p)}. \quad (3)$$

**Credentials and wages.** A degree programme in regime  $r \in \{\text{pre}, \text{post}\}$  is a menu  $(L_{fr}, T_{fr})$ : the first credential is awarded after year  $L_{fr}$  and the full degree after year  $T_{fr}$ . The reform gave STEM (engineering, the sciences and computing) a three-year exit inside its five-year degree, from  $(-, 5)$  to  $(3, 5)$ ; moved every non-STEM credential a year earlier, from  $(4, 6)$  to  $(3, 5)$ ; and left Nursing at  $(-, 4)$  (Section 2). A student who leaves after  $t$  years earns, every year until retirement, the outside wage  $W_0$  if she holds no credential, the first-degree wage  $W_f^L(p)$  if  $t \geq L_{fr}$ , and the full-degree wage  $W_f^M(p)$  if  $t = T_{fr}$ , where  $p$  is her belief

when she leaves. Wages are linear in the belief,

$$W_f^j(p) = a_f^j + g_f^j p, \quad j \in \{L, M\}, \quad (4)$$

so  $a_f^j$  is the wage of a known bad match and  $a_f^j + g_f^j$  that of a known good match. A year of study costs  $k_f$ , in units of the outside wage. A student who starts work after  $t$  years is paid for  $R - t$  years, where  $R$  is a common retirement date, so the present value of one unit a year from  $t$  on is  $A(t) = (1 - \beta^{R-t})/(1 - \beta)$ . Each extra year of study therefore costs a year of earnings.

## 6.2 The student's problem

Let  $V_{ft}(p)$  be the value of a student in field  $f$  after  $t$  completed years with belief  $p$ . She compares leaving, with the best credential available, to studying one more year:

$$V_{ft}(p) = \max \left\{ W_0 A(t), \mathbf{1}[t \geq L] W_f^L(p) A(t), \mathbf{1}[t = T] W_f^M(p) A(t), C_{ft}(p) \right\}, \quad (5)$$

$$C_{ft}(p) = -k_f + \beta \left[ q_f(p) V_{f,t+1}(p^g) + (1 - q_f(p)) V_{f,t+1}(p^b) \right], \quad t < T, \quad (6)$$

with  $C_{fT} = -\infty$ . A student who enters studies at least the first year, so at  $t = 0$  only  $C_{f0}$  is available. I solve (5) backwards from  $t = T$  on a grid of beliefs, separately for each field and regime.

**Where the reform acts.** In STEM the post-reform menu contains the pre-reform one: every strategy available before is still available, and a student who learns she is a poor match can now leave at year three with  $W^L$  instead of nothing. Hence  $V_{\text{STEM},0}^{\text{post}}(p_0) \geq V_{\text{STEM},0}^{\text{pre}}(p_0)$  for every  $p_0$ , with the largest gain where the probability of wanting to leave early is largest. This is the sense in which the reform leaves students *less to lose*: it raises the value of trying without changing the value of succeeding. In non-STEM both credentials arrive a year earlier, which saves a year of costs and earnings on every path; that gain does not depend on uncertainty. At the first credential, a student continues to the full degree if her belief is above a threshold, which is interior only if a poor match does better with the short degree and a good match with the long one (Appendix D.1).

**Field choice and outcomes.** At entry a student with prior  $p_0$  compares the values of the three fields, expressed as an equivalent yearly wage, with the outside wage, plus a field taste  $\alpha_f$  and a logit shock of scale

$\sigma$ :

$$\Pr(f \mid p_0, r) = \frac{\exp([V_{f0}^r(p_0)/A(0) + \alpha_f]/\sigma)}{\exp(W_0/\sigma) + \sum_{f'} \exp([V_{f'0}^r(p_0)/A(0) + \alpha_{f'}]/\sigma)}. \quad (7)$$

The tastes  $\alpha_f$  match the pre-reform field shares exactly and are held fixed when the reform changes the menus. Following the optimal policy forward from each  $p_0$  and integrating over the prior gives the cohort's shares in each field and on each route (drop out, first credential, full degree), and its mean wage once working,  $\bar{W}^r$ . The model's wage effect of the reform is  $\log(\bar{W}^{\text{post}}/\bar{W}^{\text{pre}})$ .

### 6.3 Estimation and fit

I fix the discount factor, the working life, the prior and the wage levels, the last from Quadros de Pessoa (Appendix D.2). I estimate seven parameters by minimum distance on the education effects in Panel A of Table 6: the taste-shock scale, the three yearly costs, the signal precisions in STEM and non-STEM, and the slope of the STEM full-degree wage. The wage effect is not targeted.

Table 6: The model and the data

| Moment                                           | Data  | (SE)   | Model | Source  |
|--------------------------------------------------|-------|--------|-------|---------|
| <i>A. Targeted</i>                               |       |        |       |         |
| $\Delta$ share of cohort in STEM (pp)            | 1.67  | (0.26) | 1.23  | QP RD   |
| $\Delta$ share in non-STEM (pp)                  | 2.03  | (0.29) | 2.59  | QP RD   |
| $\Delta$ share in Nursing (pp)                   | −0.69 | (0.14) | −0.24 | LFS RD  |
| $\Delta$ share of cohort dropping out (pp)       | −1.5  | (3.3)  | −3.5  | IEFA RD |
| $\Delta$ share with a five-year STEM degree (pp) | 1.06  | (0.19) | 0.46  | QP      |
| Non-STEM graduates with a master's, before (%)   | 18    | (3)    | 19    | QP      |
| Non-STEM graduates with a master's, after (%)    | 18    | (3)    | 17    | QP      |
| <i>B. Not targeted</i>                           |       |        |       |         |
| Wage effect, whole cohort (log points)           | 3     |        | 2.5   | QP RD   |

*Notes:* RD effects at the 2007 cutoff, cohorts 2000–2012, 2006 excluded, both sexes. Field shares: in the data, the share of the cohort holding a degree in the field; in the model, the share entering it (Appendix D.2). Dropout levels: entrants of cohorts 2000–2005 and 2007–2012 in the 2016 and 2022 IEFA waves, ages 24 and above. Source: `calibration_step2_pisa_split_dchange_moments.csv`.

The model reproduces the direction of every targeted response and the size of most; it falls short on the fall in Nursing and the rise in five-year STEM graduates. Its predicted wage effect, 2.5 log points, is below the estimated of 3.

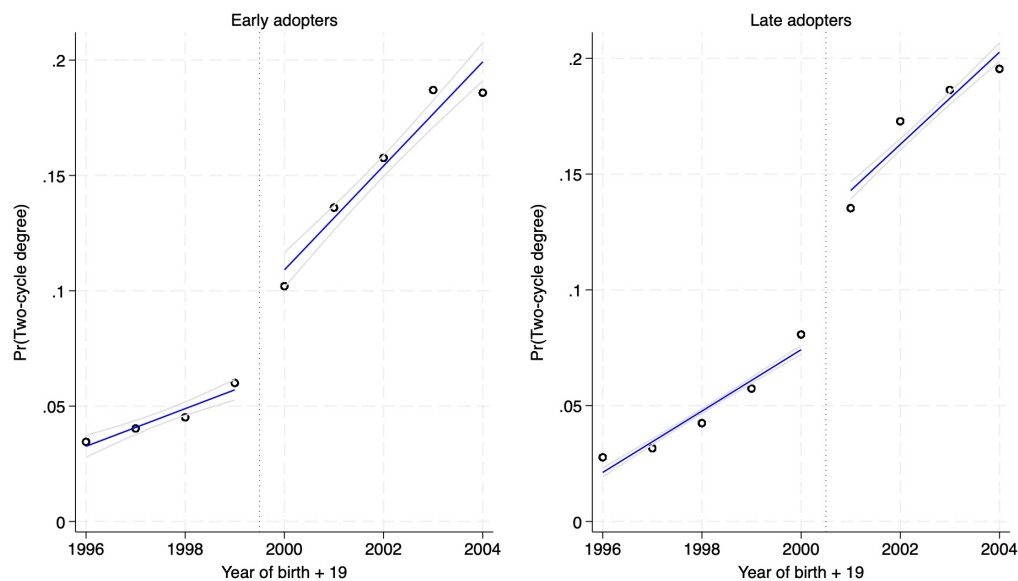
## 7 Conclusion

## A Institutional Setting: Details

### A.1 Timing of adoption

This subsection adds detail to Figure 1. In Portugal, Decree-Law 74/2006 (March 2006) let programmes convert from 2006/07 and required conversion by 2008/09. In Italy, Ministerial Decree 509/1999 took effect once the first-level degree classes were published in October 2000, with 18 months to convert. In practice, Bondonio and Berton argue that the timing of adoption was close to random, because it depended on when each faculty council first met after the decree (Bondonio and Berton, 2018). The Italian design moves the cutoff to 2000 in regions where universities adopted early, as shown in figure 10.

Figure 10: Staggered adoption of the reform in Italy



*Notes:* Italian Labour Force Survey, 2009–2020. Share of respondents whose highest degree is a two-cycle degree, by the year they turned 19, with linear fits on each side of the adoption year. Early adopters: Piemonte, Valle d’Aosta, Friuli-Venezia Giulia and Sardinia, where universities adopted the reform in 2000/01; late adopters: all other regions (2001/02). Regions are assigned by region of residence.

### A.2 What the three-year degree is worth

An earlier credential lowers the cost of a failed attempt only if employers accept it. In Portugal they were in effect obliged to, because the new degree kept the old name. The law never distinguishes licenciaturas by length or date: entry to the graduate career of the civil service requires “a licenciatura or higher degree”

(Law 12-A/2008, art. 44), and sectoral collective agreements tie wage floors to the terms “licenciado” and “bacharel” without mentioning Bologna. By default a three-year licenciado is therefore placed on the grade a four- or five-year licenciado used to occupy. Attempts to separate the two failed: the Constitutional Court struck down the bar association’s entrance exam for post-Bologna law graduates (Acórdão 3/2011), and labour courts ruled against an employer that hired only pre-Bologna licenciados. In regulated professions the value of the exit differs by field. Three-year engineering graduates can register as technical engineers, the title previously held by holders of the three-year bacharelato, and accountancy is open to any suitable licenciatura; the three-year degree carries no licence in medicine, psychology or school teaching, where a master’s became compulsory.

Italy gave the new degree a lower but still defined place. Decree 328/2001 split the professional registers of engineers, architects, biologists, chemists, geologists and psychologists into a section for master’s graduates and a “junior” section open to three-year graduates, and the three-year degree gives access to graduate grades in local government. Unlike in Portugal, however, the old four-to-six-year laurea was formally equated with the new master’s, not with the three-year degree, for public-sector competitions. The pooling of old and new credentials that operates in Portugal is therefore weaker in Italy.

### **A.3 Other changes at the same time**

*Conversion of enrolled students.* In both countries students already enrolled could convert to the new degrees, with credit for the exams they had passed. Cohorts just before each cutoff could therefore still leave with the new credential, so estimates for completion and later outcomes are, if anything, understated.

*Entry caps and nursing (Italy).* Law 264/1999 capped entry nationally to medicine, veterinary medicine, dentistry, architecture, the health professions and primary-teacher education, so entry in those fields reflects the number of places set each year. Nursing entry grew throughout 1997–2010 as training moved from regional schools into universities; the reform renamed the nursing diploma a first degree but did not shorten it.

*Fees.* Tuition fees did not change at either cutoff.

## B Data: Details

### B.1 Variable definitions

*Educational attainment.*—The main education outcome is an indicator for holding a university degree as the highest completed level. In QP this covers the *bacharelato*, *licenciatura*, *mestrado* and doctorate; once a worker is observed with a degree, the indicator stays on in later years. In the Portuguese surveys it covers the corresponding higher-education codes, and in Italy any tertiary title. In Portugal, the split between *licenciatura* and *mestrado* is only available in the data for some years, hence I refrain from using this distinction and use year of education proxies where possible. In Italy the corresponding variables are a two-cycle degree, which measures take-up of the new structure, and a degree of four or more years (a single-cycle degree or a *laurea specialistica*).

*Years of university.*—In the surveys, years of university equal the age at which the respondent left education minus 18 (19 in Italy) for graduates, and zero for everyone else. QP has no such question. I proxy it with the age at which a worker is first observed with their highest degree, minus 18; because this also counts any delay before the first private-sector job, I use it only as a proxy to be validated using the surveys.

*Field of study.*—I group fields by what the reform did to the length of the first credential (Section 2). In QP the field comes from the last two digits of the education code and allow me to separate between STEM, where a five-year programme gained a three-year exit; economics and management, the social sciences and the humanities, where a four-year programme became three plus two; and fields with no change, namely health and law. The Portuguese surveys use the broad fields of the national classification (CNAEF), in which law cannot be separated from the social sciences and business. In Italy I group degree subjects into STEM, humanities and teaching, the social sciences (including law and psychology), and health.

*Occupation and job level.*—I classify occupations into managers, professionals, technicians, clerical workers, craft workers and low-skill occupations (service and sales workers, operators and elementary occupations). QP switches from the CNP-94 to the CPP-2010 classification in 2010. I map each CNP-94 code to its most common CPP-2010 major group using INE’s correspondence table, which matches 98.9 percent of worker-years. Separately, I use the worker’s hierarchical level within the firm, whose top two levels are senior

and middle management (*quadros*). This is a job level, not an occupation. In Italy occupations follow the CP2011 major groups.

*Wages and earnings.*— The QP hourly wage is monthly October pay divided by normal monthly hours, deflated to 2010 euros with the consumer price index. Earnings and hours are each capped at their 99th percentile within year before the ratio is taken. Wages compared at a fixed age need one further step. At a fixed age the cohort index is the calendar year, so pre-reform cohorts reach a given age during the 2011–14 crisis and post-reform cohorts during the recovery and the minimum-wage increases after 2015. Year fixed effects cannot remove this, because at a fixed age the year is an exact linear function of the running variable. I therefore subtract from each log wage a calendar index estimated outside the cohort panel. The index is the mean log real hourly wage of workers aged 40–45 in each year, computed within cells of sex, education and NUTS2 region and weighted by each cell’s 2005 share of workers, so it moves with the wage level and not with who is employed. It is normalised to zero in 2005.

I measure cumulative real earnings to a given age by summing annual earnings, set to zero in years the worker is not in QP. I pad each worker’s record to one row per age from school-leaving age to age  $A$ , set annual earnings to zero where no earnings are observed, and sum. Zeros capture delayed entry; the measure does not condition on employment.

To separate pay set by the job from pay set by the employer, I define the floor of a job title as the most common log hourly wage within job title and year (in cells of at least ten workers) and the cushion as the gap between a worker’s log wage and that floor. In the surveys employment follows the ILO definition and pay is set to zero for those not employed.

*Enrolment and dropout.*—From IEFA I define entry to higher education as having completed a degree or having started one and left, and dropout as having started and left without a degree. I report dropout as a share of the whole cohort.

Table 7: Summary statistics, cohorts before and after the reform

| Pre-reform cohorts                             | Post-reform cohorts |
|------------------------------------------------|---------------------|
| <i>Panel A. Quadros de Pessoal</i>             |                     |
| <i>Panel B. Portuguese Labour Force Survey</i> |                     |
| <i>Panel C. Education and Training Survey</i>  |                     |
| <i>Panel D. Italian Labour Force Survey</i>    |                     |

*Notes:* Means. Portugal: pre-reform cohorts turned 18 in 2000–2005 and post-reform cohorts in 2007–2012; the 2006 cohort is excluded. Italy: cohorts within four years of the regional adoption year, citizens aged 30–36 at interview. Raw means mix the reform with cohort trends and with differences in the age and year of observation, and are not estimates of the reform’s effect. Hourly wage in QP is October monthly pay divided by normal monthly hours. The Italian wage is net monthly pay in the main job, zero for those not employed and top-coded at 3,000 euros. ∙: not yet computed.

## C Empirical Strategy: Details

### C.1 What the estimates measure

The coefficient  $\alpha_1$  measures the effect of facing the reformed degree structure at the age of entry, averaged over the whole cohort. The average includes people who never enrol, because they are part of the population whose choices the reform could change. I do not rescale  $\alpha_1$  by a first stage. The reform did not offer a treatment that some students took up and others refused; it changed the options open to every student, both the decision to enrol and the decision of when to stop.

### C.2 Validity of the design

**Density of the running variable** Birth year cannot be manipulated, but the number of workers per cohort in *Quadros de Pessoal* could still jump at the cutoff if the reform changed who enters the private sector. I test for a break in the density of the running variable in the spirit of McCrary (2008). For each cohort bin  $j$  I compute  $\hat{f}(z_j) = N_j/(Nh)$ , where  $N_j$  is the number of workers in the bin,  $N$  the total and  $h = 1$  the bin width, and estimate

$$\log \hat{f}(z_j) = \gamma_L + \delta_L z_j + u_j \quad (z_j < 0), \quad \log \hat{f}(z_j) = \gamma_R + \delta_R z_j + v_j \quad (z_j \geq 0), \quad (8)$$

weighting each bin by  $N_j$  and using a triangular kernel within a symmetric window around the cutoff. The discontinuity in the log density is  $\hat{\gamma}_R - \hat{\gamma}_L$ . Figure 11 shows no visible break. The estimated jump in the log density is 0.019, with a standard error of 0.020.

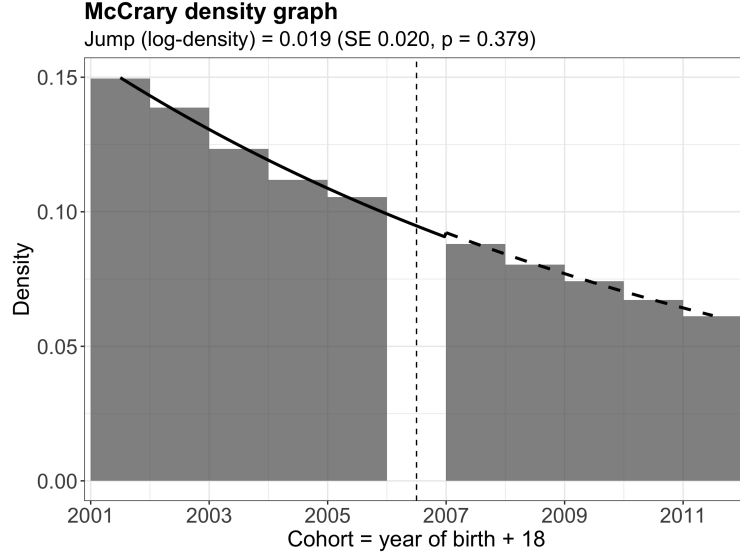


Figure 11: Density of the running variable in *Quadros de Pessoal*

*Notes: Quadros de Pessoal.* Share of workers by the year they turned 18, with fits from equation (8) on each side of the cutoff; the dotted line marks the excluded 2006 cohort. The estimate at the top of the panel is the jump in the log density at the cutoff.

**No delayed entry** Students who turned 18 shortly before 2007, and who expected the reform, could have postponed entry to enrol in the shorter degrees. Such students would then be counted as untreated although they studied under the new structure, and the pre-reform cohorts closest to the cutoff would be selected. Postponing entry usually means working in the meantime, so I estimate equation (2) with an indicator for being observed in employment between leaving secondary school and completing a university degree. Table 8 shows that the discontinuity is close to zero and precisely estimated for women and men alike.

Table 8: Employment between secondary school and university

|                      | Whole sample     | Women            | Men               |
|----------------------|------------------|------------------|-------------------|
| RD jump              | 0.001<br>(0.001) | 0.002<br>(0.002) | −0.000<br>(0.002) |
| Baseline             | 0.105            | 0.128            | 0.085             |
| p-value, women = men | 0.283            |                  |                   |

*Notes: Quadros de Pessoal.* The outcome is an indicator for being observed in employment between leaving secondary school and completing a university degree. Estimates of equation (2). Standard errors in parentheses. \* p<0.10, \*\* p<0.05, \*\*\* p<0.01.

### C.3 AKM decomposition of wages

A wage gain can arise because workers from reformed cohorts are paid more in the same firms or because they work in firms that pay all their workers more. To separate the two I estimate a two-way fixed-effects wage equation (Abowd et al., 1999),

$$\log w_{it} = \theta_i + \psi_{J(i,t)} + \mu_t + h(a_{it}) + \epsilon_{it}, \quad (9)$$

where  $\theta_i$  is a worker effect,  $\psi_{J(i,t)}$  the pay premium of the firm  $J(i, t)$  that employs worker  $i$  in year  $t$ ,  $\mu_t$  a year effect and  $h(\cdot)$  a quadratic in age. Worker and firm effects are separately identified only among firms linked by workers who move between them, so I estimate equation (9) on the largest connected set of firms and workers.

I then run the discontinuity regression on the wage and on its firm component. Let  $\tilde{w}_{it}$  be the log hourly wage after removing a quadratic in age and year effects. Because  $\tilde{w}_{it} = \hat{\psi}_{J(i,t)} + (\tilde{w}_{it} - \hat{\psi}_{J(i,t)})$ , and because the discontinuity regression is linear and is run on the same worker-years for each term, the wage jump splits exactly into two parts:

$$\Delta^w = \Delta^\psi + \Delta^{w-\psi}, \quad \text{firm share} = \frac{\Delta^\psi}{\Delta^w}. \quad (10)$$

The firm component  $\Delta^\psi$  measures how much of the wage jump comes from working in firms with higher pay premia. The remainder  $\Delta^{w-\psi}$  contains the worker effects and everything else. Firm premia are held fixed over the sample period, so the firm share captures moves towards firms that pay more to all their workers, not changes in how firms pay.

### C.4 Firm sector decomposition of wages

The AKM decomposition separates the wage effect into worker and firm components, but it does not say in which firms the gain is realised. I therefore split the effect on the log wage by the type of firm the worker is employed in. The split is exact, and each of its pieces is an RD on an outcome defined for every worker, so no piece conditions on where the worker ends up.

**Firm types.** I classify each employer by its three-digit CAE Rev. 3 sector. QP codes some firms at division level in some years and at group level in others (banks move between 641 and 640, insurers between

651 and 650), so for those divisions I use the division code in every year. I group sectors into four types. *IT and engineering*: computer programming and IT consulting (620), data processing and hosting (631), software publishing (582), computers and electronics (26) and motor-vehicle parts (293). *Business and support services*: head offices (701), management consultancy (702), accounting, audit and tax (692), other professional and technical services (749), and divisions 77–82 (rental, staffing, travel, security, cleaning, office administration and business support). *Banking, insurance and finance*: divisions 64–66. *All other firms*.

**Outcomes.** Let  $w_i$  be the calendar-deflated log hourly wage of worker  $i$ ,  $k(i)$  the type of the firm they work in, and  $c$  the mean of  $w$  among pre-reform cohorts in the estimation window. For each type  $k$  I define

$$y_{ik} = (w_i - c) \mathbf{1}\{k(i) = k\}, \quad (11)$$

which is zero for everyone not employed in a type- $k$  firm. Summing over types gives  $\sum_k y_{ik} = w_i - c$ . The RD estimator is linear in the cohort cell means and  $c$  is a constant, so the RD estimates on the  $y_k$  add up exactly to the RD on  $w$ . The mean of  $y_k$  in a cohort is  $s_k(\mu_k - c)$ , where  $s_k$  is the share of the cohort working in type  $k$  and  $\mu_k$  is their mean log wage. Its jump at the cutoff is

$$\Delta[s_k(\mu_k - c)] = \underbrace{\Delta s_k (\mu_k^- - c)}_{\text{more workers in } k} + \underbrace{s_k^- \Delta \mu_k}_{\text{higher pay inside } k} + \underbrace{\Delta s_k \Delta \mu_k}_{\text{interaction}}, \quad (12)$$

where a superscript  $-$  denotes the limit from the pre-reform side. A firm type therefore adds to the wage effect if post-reform cohorts pay more inside it, if more of them work there and it paid above the pre-reform average, or if fewer of them work there and it paid below that average.

The rows say where the reform’s wage effect is realised, not how large it would have been had workers not changed sector. Sector is itself an outcome of the reform, so a comparison of wages *within* a sector would compare post-reform workers with pre-reform workers who would have chosen differently. The outcomes  $y_k$  avoid that comparison because they are defined for everyone.

## C.5 Wage floor and wage cushion decomposition

Following [Cardoso and Portugal \(2005\)](#), I split each worker’s wage into the part set by the job title and the part set by the employer. Most Portuguese private-sector workers are covered by collective agreements that

fix a minimum rate for each job title (*categoria profissional*). Firms may pay more than that rate but not less. The wage is therefore the sum of a *floor*, the rate attached to the worker’s job title, and a *cushion*, the pay the firm adds above it.

**Measuring the floor.** I do not observe the agreed rates directly, so I recover them from the distribution of pay. Because many workers are paid exactly the rate of their job title, that rate is where wages within the title bunch. For job title  $c$  in year  $y$  I define the floor  $F(c, y)$  as the most common log hourly wage, rounded to 0.01 log points, among all worker-years in *Quadros de Pessoal* that hold the title in that year. Ties go to the lower wage. I use every worker-year in the data, not only the cohorts in the estimation sample, because the floor belongs to the job title rather than to the worker. I compute it only in cells with at least ten worker-years, and I leave out the residual code for workers with no professional category. At age 30, [TK: 80] percent of workers in the sample of Table 2 hold a job title with a measured floor.

**The decomposition.** Let  $w_{it}$  be the log hourly wage of worker  $i$  in year  $t$ ,  $c(i, t)$  their job title and  $I(t)$  the calendar index of Appendix B.1. Then

$$w_{it} - I(t) = \underbrace{[F(c(i, t), t) - I(t)]}_{\text{floor}} + \underbrace{[w_{it} - F(c(i, t), t)]}_{\text{cushion}}. \quad (13)$$

As in the other two splits, the discontinuity regression is linear and is run on the same worker-years for each term, so the wage jump is exactly the sum of the jump in the floor and the jump in the cushion,  $\Delta^w = \Delta^F + \Delta^{w-F}$  (Table 4). The floor is computed from the unwinsorised wage. The cushion is measured against the winsorised wage, so the two terms add up to the wage in Table 2.

## D The model: details

### D.1 The continuation threshold

At the year of the first credential a student with belief  $p$  chooses between leaving with  $W_f^L(p)$  and continuing. Ignoring the value of the further signals she would receive, continuing for  $T - L$  more years is worth it when

$$\beta^{T-L} A(T) W_f^M(p) - k_f \sum_{s=0}^{T-L-1} \beta^s \geq A(L) W_f^L(p). \quad (14)$$

With the linear wages (4), the left side grows faster in  $p$  than the right whenever  $\beta^{T-L}A(T)g_f^M > A(L)g_f^L$ , so (14) defines a threshold belief above which students continue. Signals received while continuing lower the threshold further, because a student who continues can still leave with  $W_f^L$  if they turn out bad. The threshold is interior only if the two wage lines cross; in the calibration they cross at  $p = 0.31$  in STEM and  $p = 0.27$  in non-STEM (Table 9).

## D.2 Calibration and estimation

**Fixed parameters.** Table 9 lists what I set outside the model. Wages are in units of the outside wage. The full-degree wage of an average student ( $p = 0.4$ ) in each field is the ratio of the field’s mean hourly wage to that of workers without university in Quadros de Pessoa; the first-degree wage at  $p = 0.4$  is a fixed fraction of it, and its slope a fixed fraction  $\rho_f$  of the full-degree slope. The non-STEM and Nursing full-degree slopes and the Nursing signal precision are held at values from an earlier calibration.

Table 9: Fixed parameters and implied wages

|                                                   | STEM        | Non-STEM    | Nursing     |
|---------------------------------------------------|-------------|-------------|-------------|
| Full-degree wage at $p = 0.4$ (outside = 1)       | 1.98        | 1.62        | 1.69        |
| First-degree wage at $p = 0.4$ , relative to full | 0.911       | 0.925       | –           |
| Slope ratio $\rho_f = g_f^L/g_f^M$                | 0.6         | 0.5         | –           |
| Full-degree slope $g_f^M$                         | estimated   | 0.73        | 0.64        |
| Signal precision $\pi_f$                          | estimated   | estimated   | 0.29        |
| <i>Implied wages, bad match / good match</i>      |             |             |             |
| First degree                                      | 1.25 / 2.18 | 1.28 / 1.65 | –           |
| Full degree                                       | 1.05 / 2.60 | 1.18 / 1.91 | 1.31 / 1.95 |
| Belief at which the two lines cross               | 0.31        | 0.27        | –           |

*Notes:* Common to all fields:  $\beta = 0.95$ , retirement 38 years after entry,  $W_0 = 1$ , prior Beta(1.22, 1.83), 301 grid points. First-degree ratios are the licenciatura wage over the field mean wage (7.75/8.51 and 6.43/6.95). STEM implied wages use the estimated slope  $g_{\text{STEM}}^M = 1.55$ .

**Estimator and identification.** I minimise  $\sum_m [(\hat{m}_m - m_m(\theta))/se_m]^2$  with Nelder–Mead from five starting points, with bounds imposed by a logistic transformation. The two best starting points reach losses of 27.7 and 28.7 near the same parameters; none ends at a bound. The taste scale  $\sigma$  is pinned by the size of the share responses. The non-STEM cost and precision are pinned by non-STEM continuation before and after the reform. The three STEM parameters move the five-year STEM share in offsetting directions, so each is imprecise alone (Table 10); the data determine their combination, the net value of the full degree to

a student who has learned she is a good match. The Nursing cost is not identified, since the Nursing menu does not change and its cost is absorbed by the Nursing taste; it affects no targeted moment.

Table 10: Estimated parameters

| Parameter                                          | Estimate | (SE)           |
|----------------------------------------------------|----------|----------------|
| Taste-shock scale $\sigma$                         | 0.323    | (0.046)        |
| Yearly cost, STEM $k_{\text{STEM}}$                | 0.020    | (0.353)        |
| Yearly cost, non-STEM $k_{\text{non-STEM}}$        | 0.028    | (0.002)        |
| Yearly cost, Nursing $k_{\text{Nursing}}$          | 0.052    | not identified |
| Signal precision, STEM $\pi_{\text{STEM}}$         | 0.886    | (12.3)         |
| Signal precision, non-STEM $\pi_{\text{non-STEM}}$ | 0.355    | (0.240)        |
| Full-degree slope, STEM $g_{\text{STEM}}^M$        | 1.551    | (9.75)         |

*Notes:* Standard errors from  $(G'WG)^{-1}$ , with  $G$  the Jacobian of the targeted moments at the estimate (central differences) and  $W$  the inverse of the moments' variances; local and approximate, since the policy functions have kinks on the belief grid. Source: `section.parameters.step2.pisa.split.dchange.csv`.

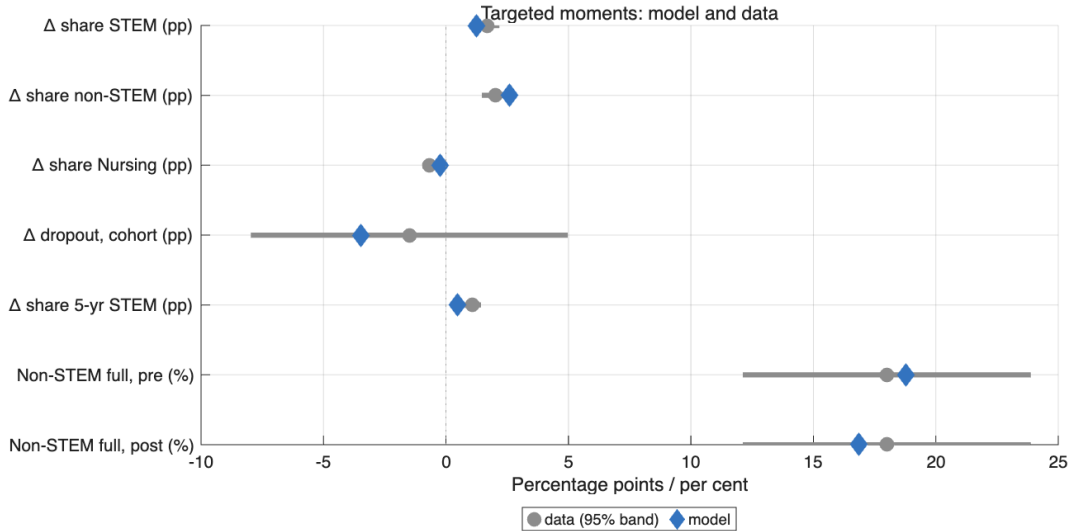


Figure 12: Targeted moments: model and data

*Notes:* Grey: data with 95 percent intervals; blue: model. Percentage points for changes, percent for continuation shares. The same numbers as Table 6, Panel A.

### D.3 Decomposing the wage effect

Table 11 splits the model's wage effect by the route students end up on. The effect comes from where students end up, not from how well they are matched: within each route the mean wage barely changes. The largest contribution, about one log point, comes from students leaving STEM with the new three-year degree, and 0.8 of it from students who would have dropped out of STEM before the reform. Non-STEM adds about one log point, mostly through more first-degree graduates; Nursing subtracts a little.

Table 11: Where the model’s wage effect comes from

| Route                         | Share of cohort (%) |       | Wage<br>(outside = 1) | Contribution<br>(% of mean wage) |
|-------------------------------|---------------------|-------|-----------------------|----------------------------------|
|                               | before              | after |                       |                                  |
| Never enter                   | 74.0                | 70.5  | 1.00                  | 0                                |
| STEM, drop out                | 3.5                 | 0.0   | 1.00                  | 0                                |
| STEM, three-year degree       | 0.0                 | 4.3   | 1.26                  | 0.95                             |
| of which former STEM dropouts |                     |       |                       | 0.78                             |
| STEM, five-year degree        | 4.3                 | 4.8   | 2.58                  | 0.72                             |
| Non-STEM, first degree        | 11.0                | 13.5  | 1.39                  | 0.89                             |
| Non-STEM, full degree         | 2.6                 | 2.7   | 1.87                  | 0.11                             |
| Nursing                       | 4.6                 | 4.3   | 1.60                  | −0.13                            |
| Total                         |                     |       |                       | 2.54                             |

*Notes:* Contribution of route  $j$ :  $[s_j^{\text{post}}(w_j^{\text{post}} - 1) - s_j^{\text{pre}}(w_j^{\text{pre}} - 1)]/\bar{W}^{\text{pre}}$ , with  $\bar{W}^{\text{pre}} = 1.161$ ; the total is the percent change in the cohort mean wage (2.51 log points). Wages shown are pre-reform means (post-reform for the new three-year degree).

## D.4 Counterfactuals

Each column of Table 12 holds every parameter and field taste at its estimate, changes one feature of the environment, and reports the reform’s effect within that world.

**Uncertainty.** With full information, each student knows her match at entry. The STEM response disappears, because an exit is worth something only to a student unsure whether she will want it; the non-STEM response survives, because a shorter degree saves time whether or not a student is unsure. With uncertainty but no learning, STEM entry still rises, but students can no longer learn that they are good matches, so the five-year STEM share falls by 4.4 percentage points. Uncertainty drives STEM entry; learning keeps students going to year five. Without either, the wage effect falls to about one log point.

**One field at a time.** The STEM exit alone gives a wage effect of 1.8 log points and the non-STEM change alone 0.8; each pulls students from the other field.

**Where the exit sits.** An exit at year two nearly doubles the STEM entry response, but every STEM student then takes it, and the wage effect falls to 1.3 log points. An exit at year four halves the entry response and gives 2.1. Year three gives the largest wage effect: early enough that poor matches use it, late enough that good matches have already found out.

Table 12: Counterfactuals: the reform’s effect in each world

|                                | Baseline | Uncertainty<br>full info | no learning | One field only<br>STEM | non-STEM | STEM exit at<br>year 2 | year 4 | No recog-<br>nition |
|--------------------------------|----------|--------------------------|-------------|------------------------|----------|------------------------|--------|---------------------|
| $\Delta$ entry (pp)            | 3.58     | 2.56                     | 3.67        | 1.24                   | 2.42     | 4.44                   | 3.04   | 2.96                |
| $\Delta$ share STEM (pp)       | 1.23     | −0.07                    | 1.26        | 1.57                   | −0.29    | 2.36                   | 0.53   | 0.43                |
| $\Delta$ share non-STEM (pp)   | 2.59     | 2.83                     | 2.63        | −0.24                  | 2.86     | 2.38                   | 2.71   | 2.73                |
| $\Delta$ share Nursing (pp)    | −0.24    | −0.20                    | −0.23       | −0.08                  | −0.16    | −0.29                  | −0.20  | −0.20               |
| $\Delta$ 5-year STEM (pp)      | 0.46     | −0.56                    | −4.37       | 0.65                   | −0.17    | −4.30                  | 0.22   | 0.15                |
| STEM grads, 5 years, after (%) | 53       | 85                       | 25          | 53                     | 100      | 0                      | 54     | 54                  |
| Wage effect (log pts)          | 2.51     | 1.01                     | 1.15        | 1.80                   | 0.80     | 1.33                   | 2.13   | 1.79                |

*Notes:* All parameters and field tastes at their baseline values; each column is the reform’s effect in that world. Field shares are entry shares, as in the baseline targets. Full info: match known at entry. No learning: uninformative signals. One field only: the other field keeps its pre-reform menu. No recognition: the new three-year STEM degree paid seven log points less at every belief. Source: `section.counterfactuals.step2.pisa.split.dchange.csv`.

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